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(54) **AUTOMATIC PRESENCE SIMULATOR FOR SECURITY SYSTEMS**

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**G08B 25/14** (2013.01); **H04R 2227/003**  
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See application file for complete search history.

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(63) Continuation of application No. 17/541,711, filed on Dec. 3, 2021, now Pat. No. 11,816,968, which is a (Continued)

(57)

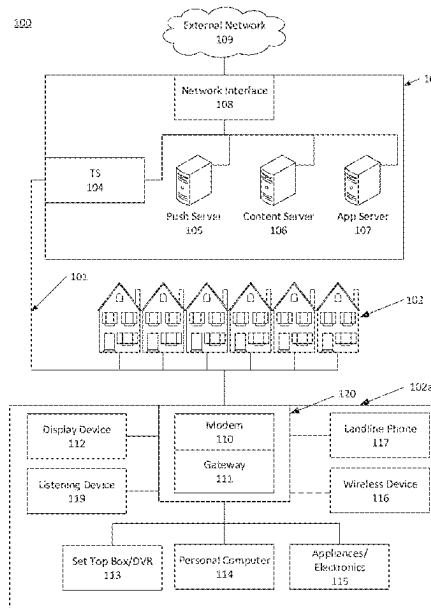
**ABSTRACT**

(51) **Int. Cl.**  
**G08B 15/00** (2006.01)  
**G05B 15/02** (2006.01)  
(Continued)

Methods, systems and computing devices are described herein for monitoring a premises when residents or other users are present in order to detect patterns of activity at the premises. Such patterns may comprise, for example, a typical schedule indicating usage of one or more devices by one or more users of the premises. When a user is away or otherwise inactive, commands may be sent to various user devices to make it appear (e.g., to those outside the premises) as if the user is present by simulating and/or recreating the patterns that were previously detected.

(52) **U.S. Cl.**  
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**17 Claims, 11 Drawing Sheets**



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## Related U.S. Application Data

continuation of application No. 16/522,841, filed on Jul. 26, 2019, now Pat. No. 11,222,516, which is a continuation of application No. 15/707,407, filed on Sep. 18, 2017, now Pat. No. 10,410,492.

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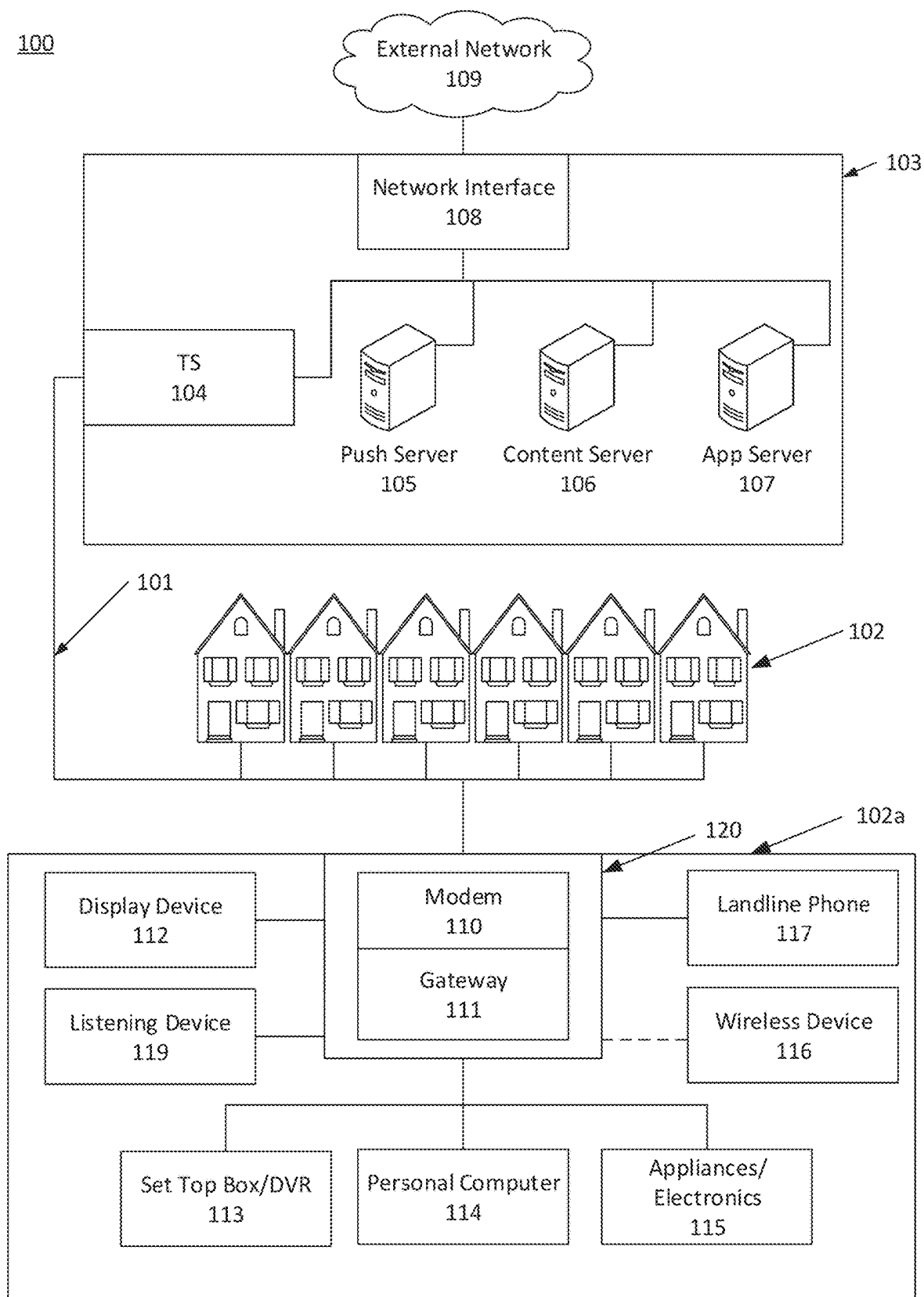


Fig. 1

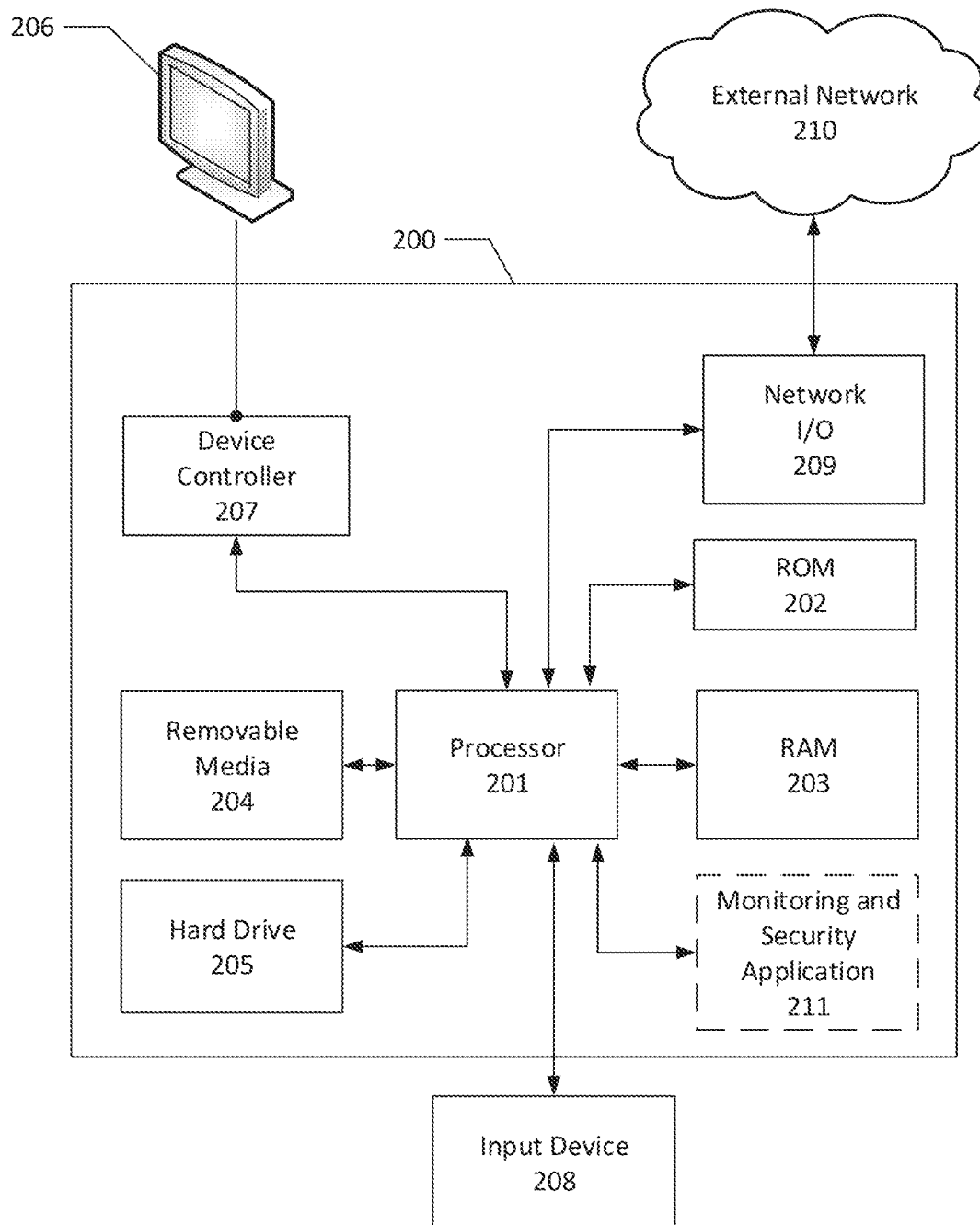


FIG. 2

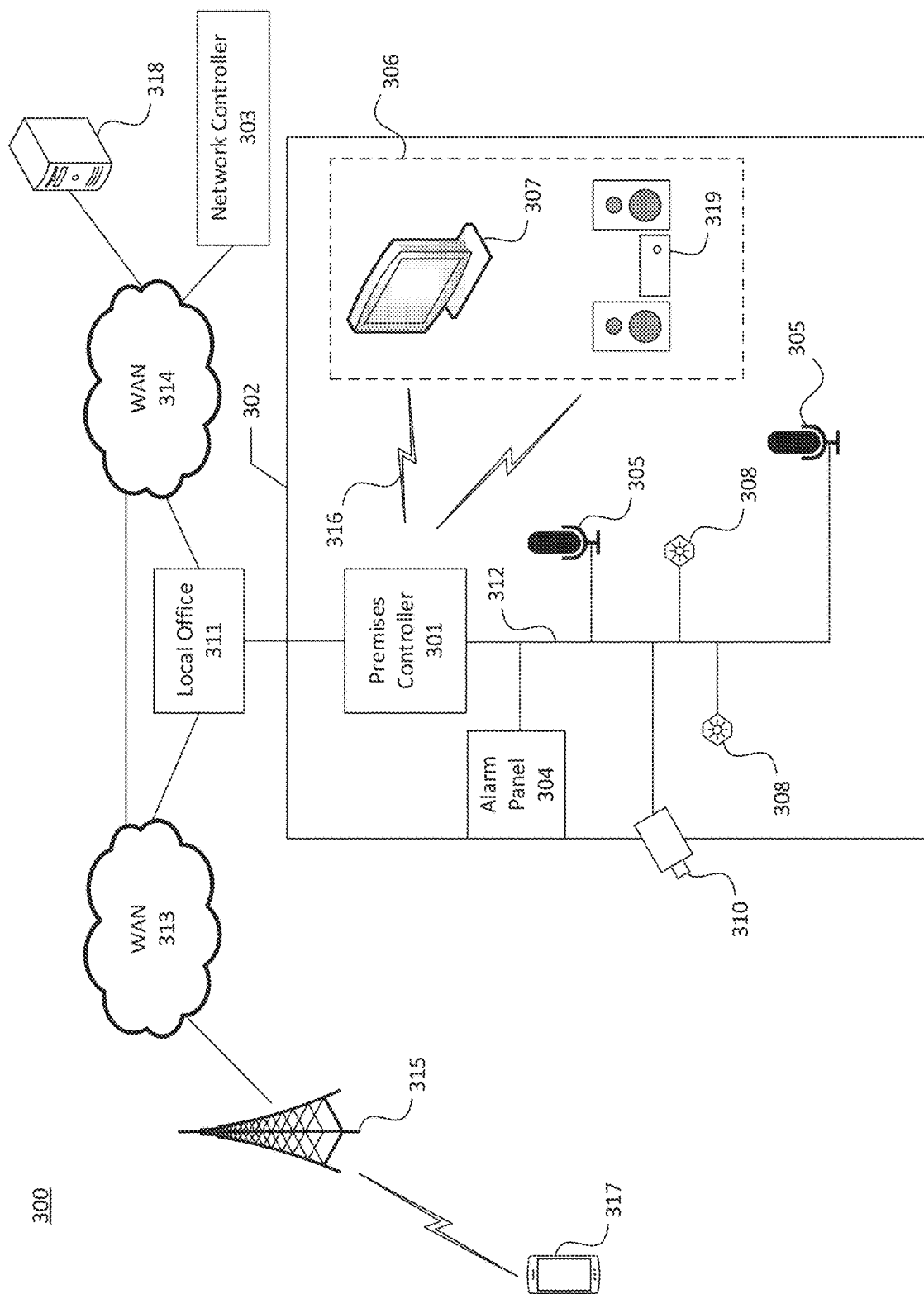
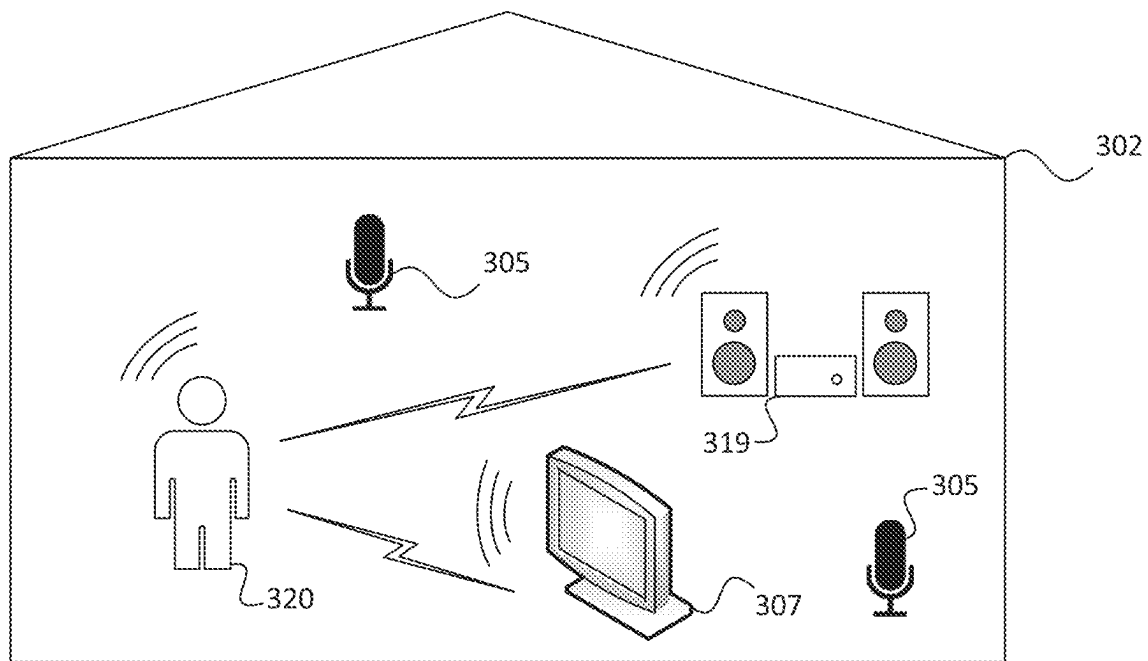
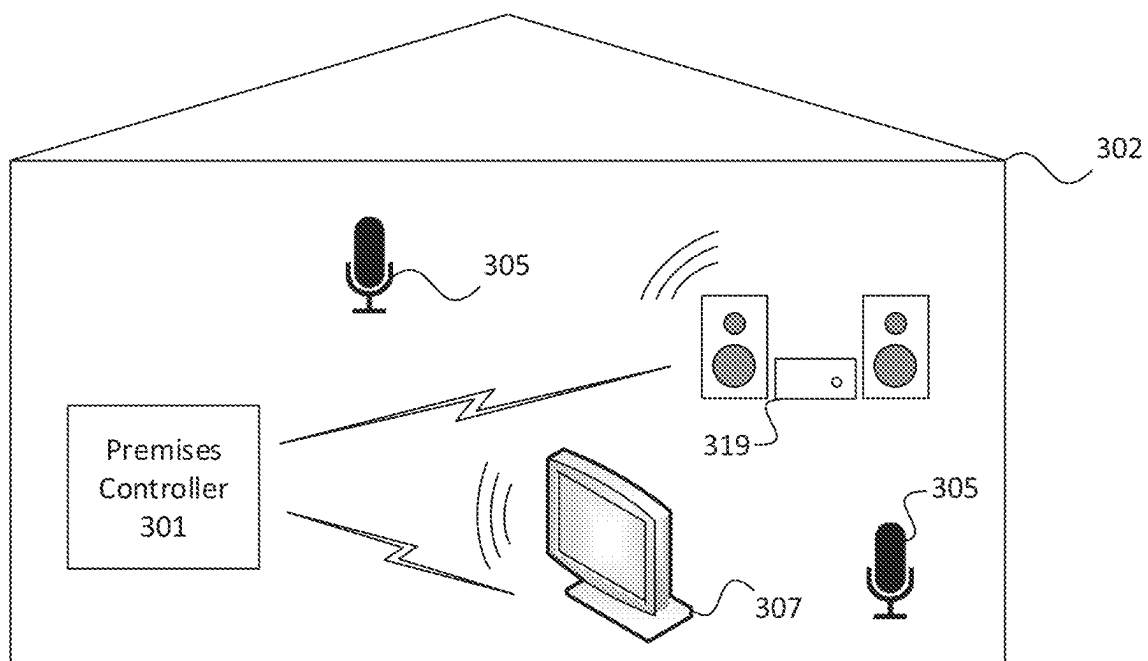


FIG. 3



A user controls devices in a premises

FIG. 4A



Premises Controller simulates a user's presence when the user is away

FIG. 4B

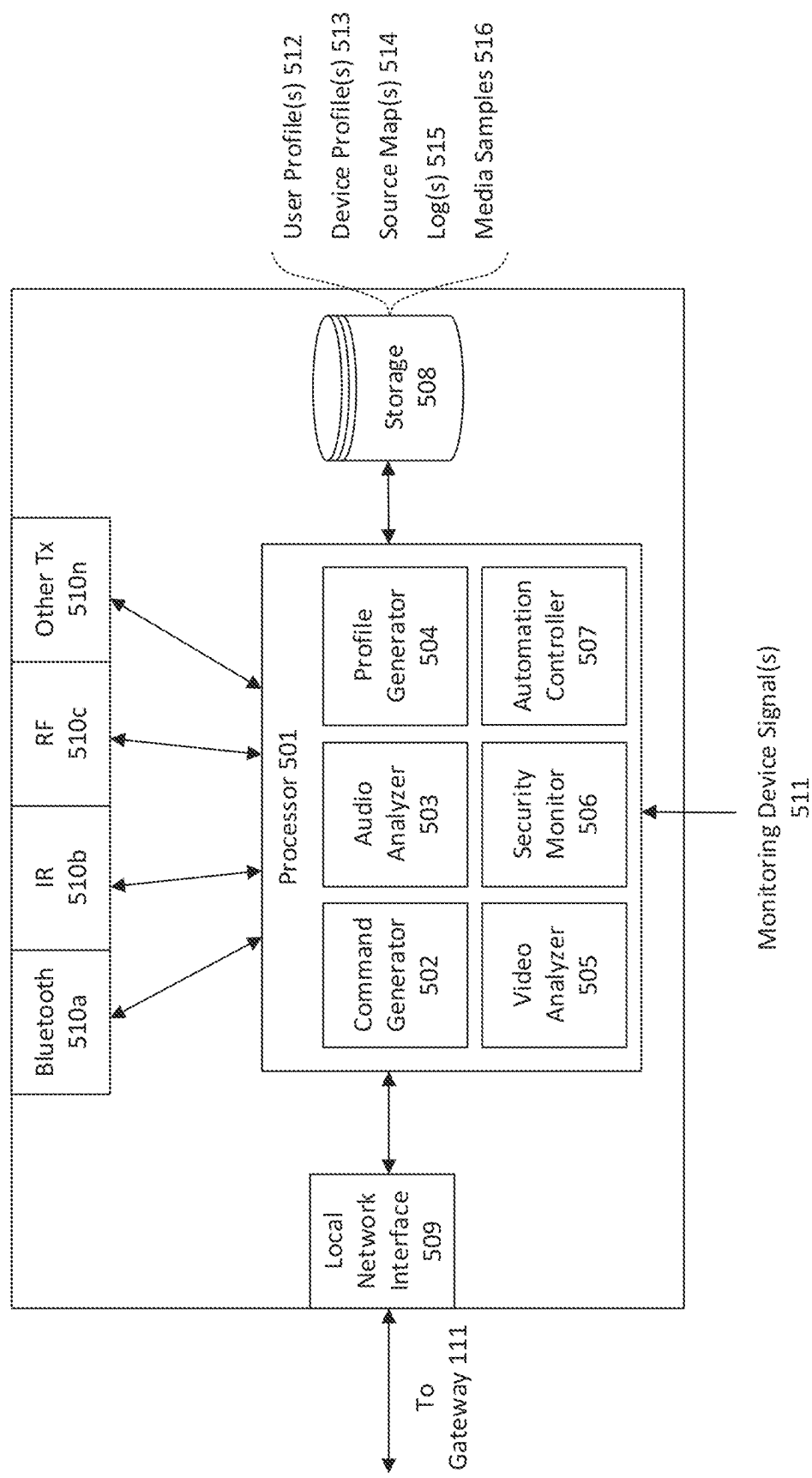


FIG. 5

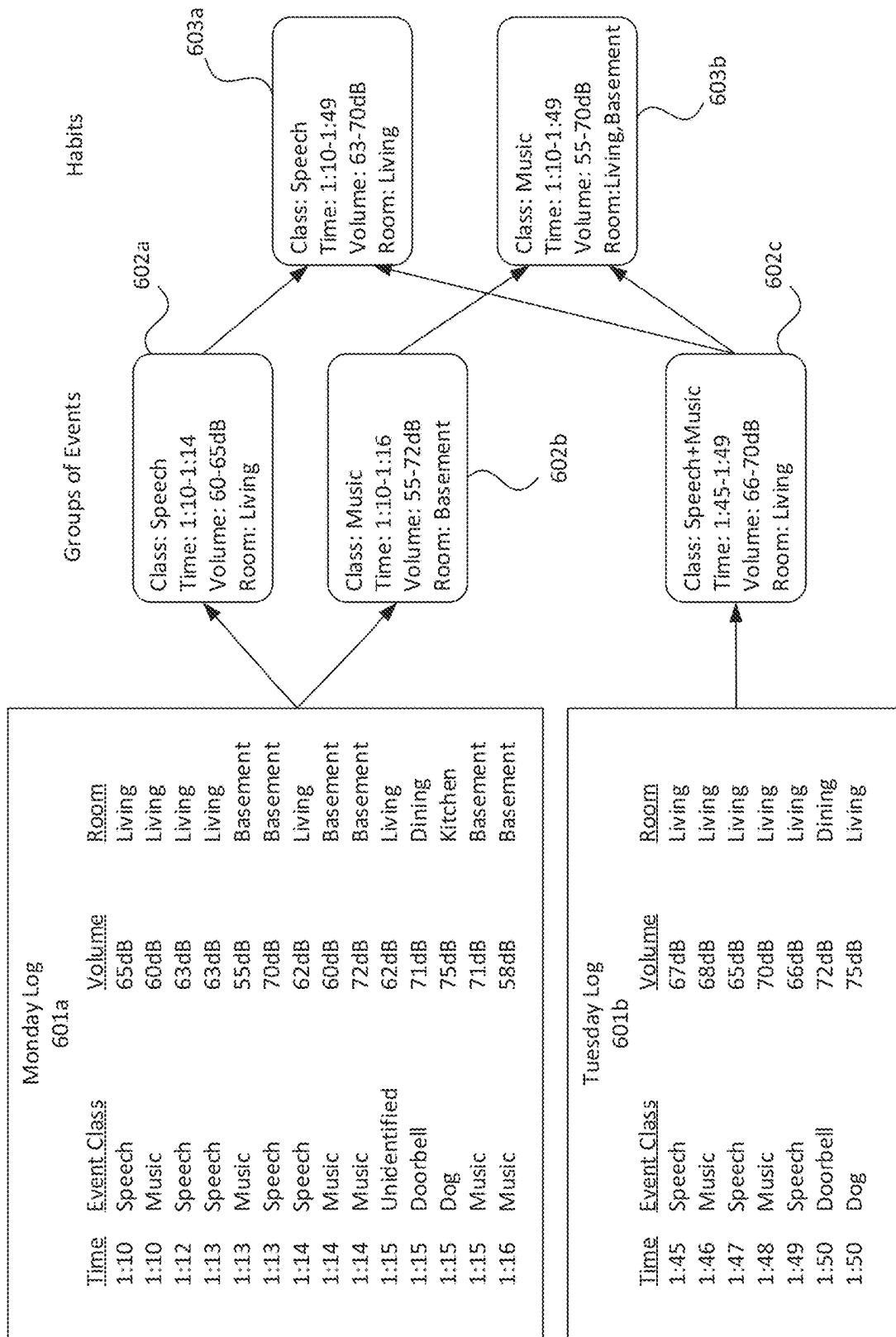


FIG. 6A



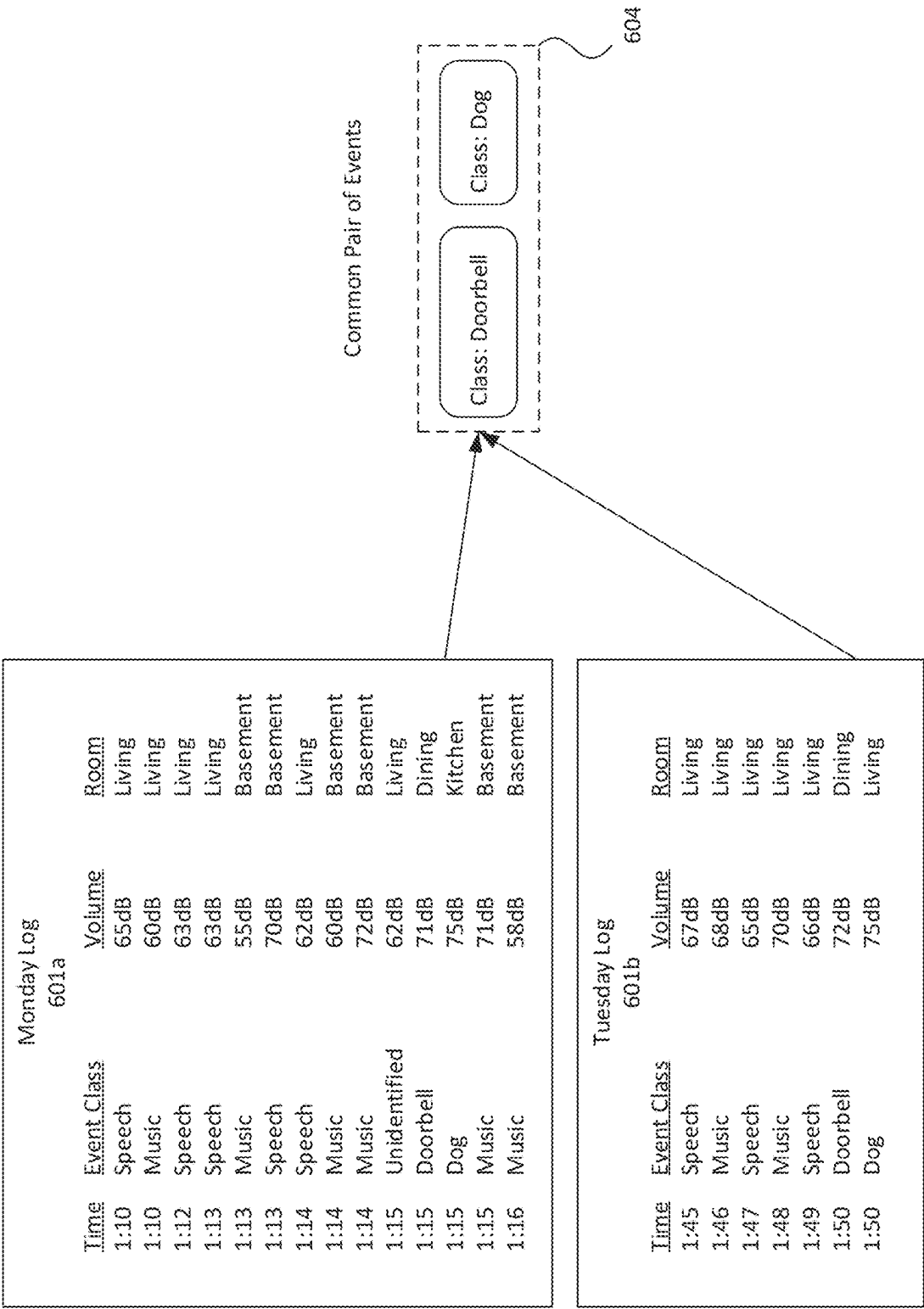


FIG. 6B

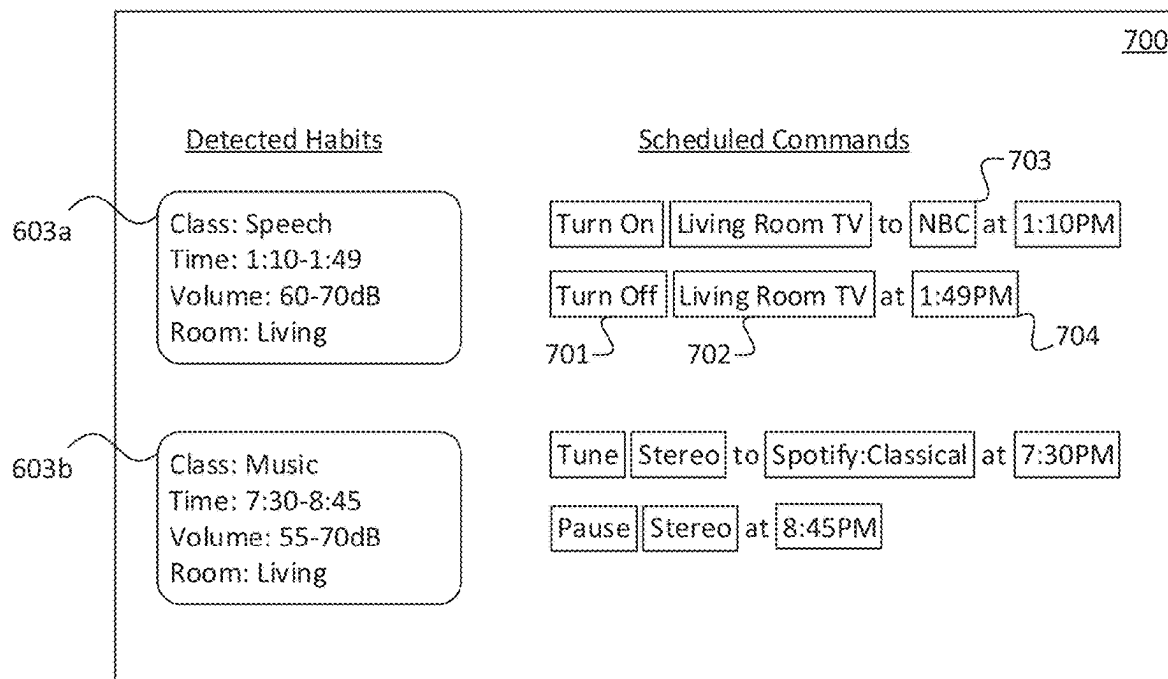


Fig. 7A

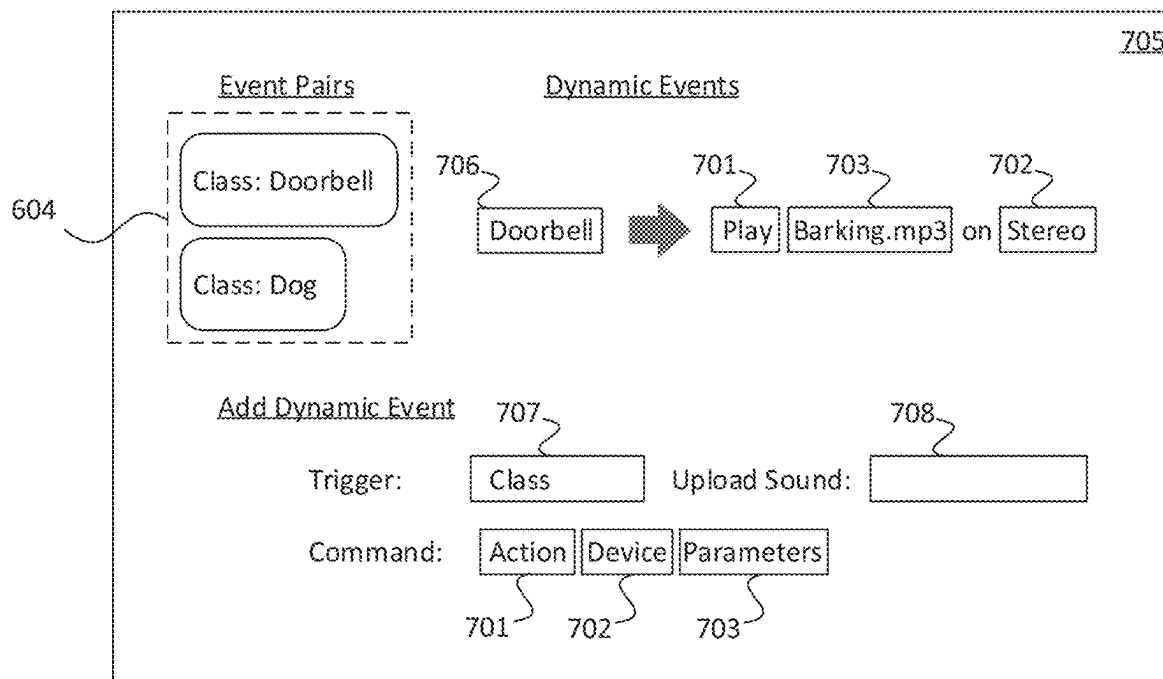


Fig. 7B

800

System detected an unidentified event on Monday at 1:15PM in the Living Room. Please review and identify the event:

Select to Play:  801

Select a Classification: 

|                                                                                                 |     |
|-------------------------------------------------------------------------------------------------|-----|
| Unidentified  |     |
| Speech                                                                                          | 35% |
| Music                                                                                           | 30% |
| Dog                                                                                             | 25% |
| Not Sure                                                                                        |     |
| Ignore                                                                                          |     |

 802

Select a Device: 

|                                                                                                   |  |
|---------------------------------------------------------------------------------------------------|--|
| Living Room TV  |  |
| Stereo                                                                                            |  |
| None                                                                                              |  |

 803

Fig. 8

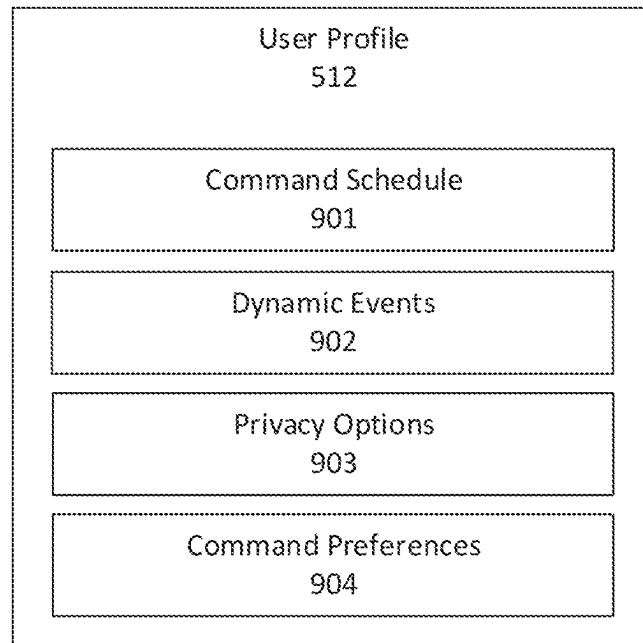


Fig. 9

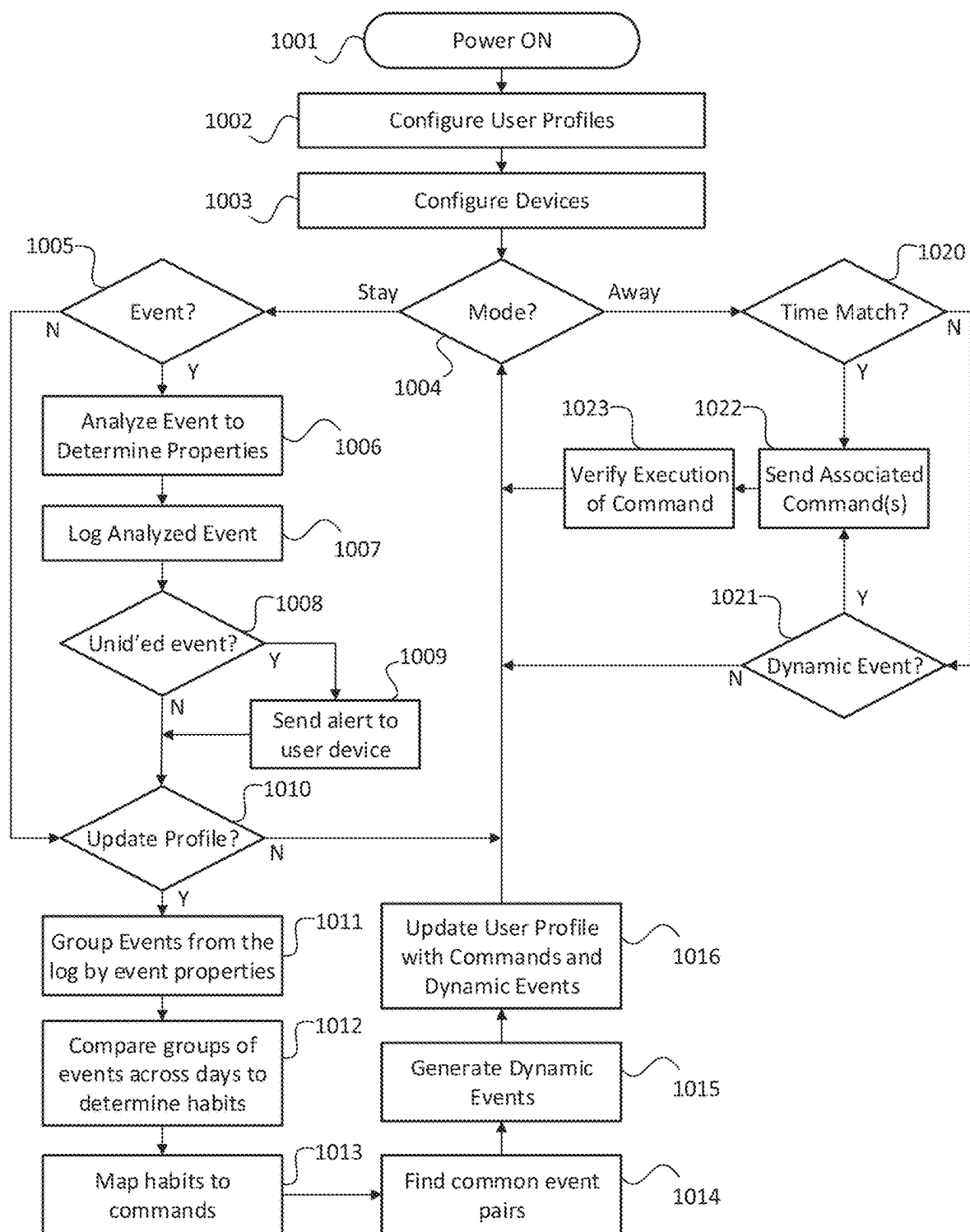


Fig. 10

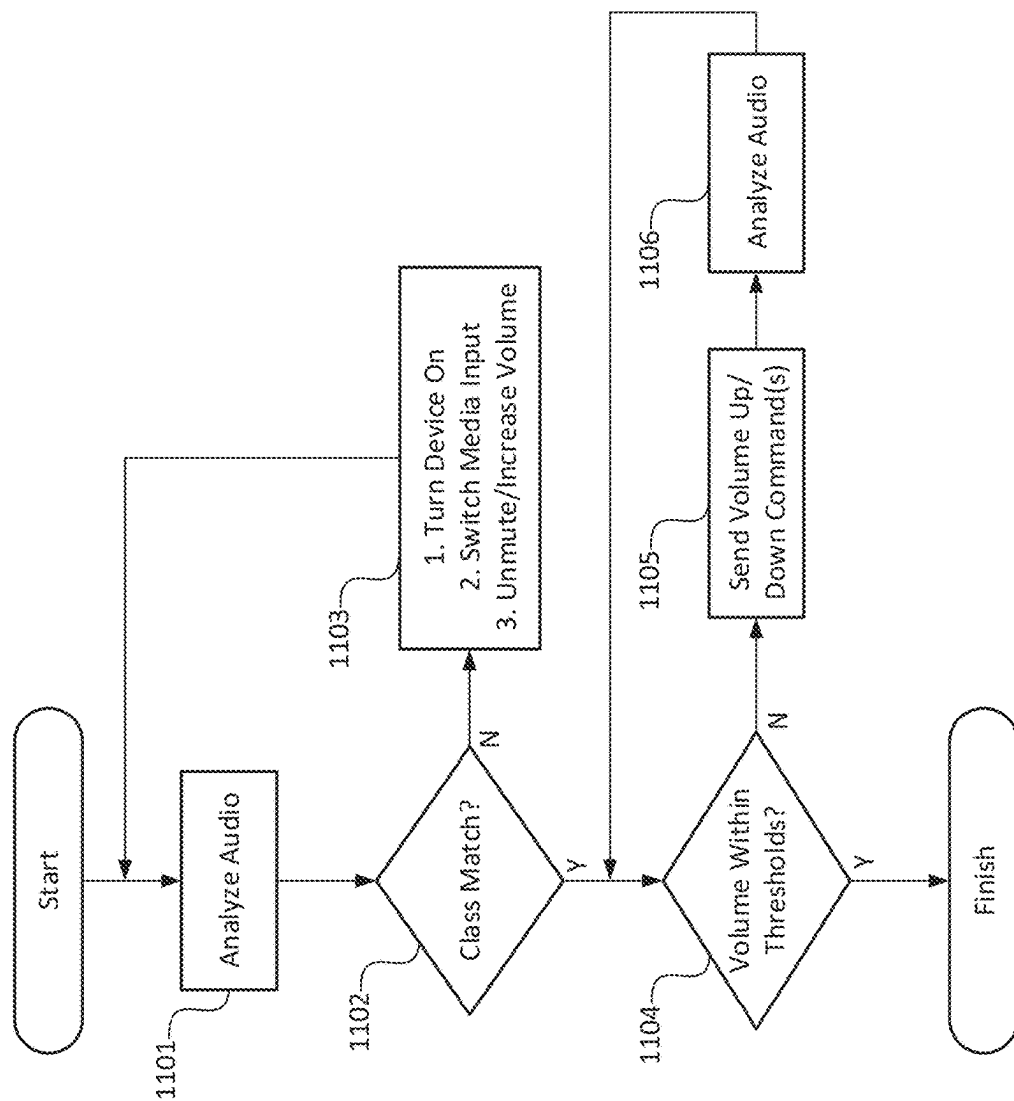


FIG. 11

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## AUTOMATIC PRESENCE SIMULATOR FOR SECURITY SYSTEMS

### RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/541,711 filed Dec. 3, 2021, which is a continuation of U.S. application Ser. No. 16/522,841 (now U.S. Pat. No. 11,222,516) filed Jul. 26, 2019, which is a continuation of U.S. application Ser. No. 15/707,407 (now U.S. Pat. No. 10,410,492) filed Sep. 18, 2017, which are entirely incorporated herein by references.

### BACKGROUND

In providing security services, security systems typically include sensors for detecting break-ins. Such sensors may detect intrusions via windows and doors while a user is away or while the security system is otherwise armed. However, there remains an ever-present need for improvements in security systems' ability to detect, and hopefully prevent, unwanted intrusions.

### SUMMARY

The following presents a simplified summary in order to provide a basic understanding of the disclosure. The summary is not an extensive overview of the disclosure. It is neither intended to identify key or critical elements of the disclosure nor to delineate the scope of the disclosure. The following summary merely presents some concepts of the disclosure in a simplified form as a prelude to the description below.

Methods, systems and computing devices are described herein for monitoring a premises when residents or other users are present in order to detect patterns of activity at the premises. The patterns may be detected from recorded audio events indicating, for example, the sounds of television programs being viewed, the turning on/off of electronic devices, use of appliances, etc. Such patterns may comprise, for example, a typical schedule indicating usage of one or more devices by one or more users of the premises. When a user is away or otherwise inactive, commands may be sent to various user devices to make it appear (e.g., to those outside the premises) as if the user is present by simulating and/or recreating the patterns that were previously detected.

The methods, systems and computing devices described herein may be included as part of a network, such as a cable television distribution network.

The details of these and other embodiments of the present disclosure are set forth in the accompanying drawings and the description below. Other features and advantages will be apparent from the description and drawings, and from the claims.

### BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure is illustrated by way of example and not limited in the accompanying figures in which like reference numerals indicate similar elements and in which:

FIG. 1 shows an example network on which the various elements described herein may be implemented.

FIG. 2 shows an example computing device on which the various elements described herein may be implemented.

FIG. 3 shows an example monitoring and security system within a premises.

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FIGS. 4A-B show an example monitoring and security system operating in different modes.

FIG. 5 shows an example premises controller for implementing the functions described herein.

FIGS. 6A-B show example data structures for storing events and generating user profiles.

FIGS. 7A-B show example user interfaces for adjusting a user profile.

FIG. 8 shows an example user interface for identifying audio signals received by the monitoring and security system.

FIG. 9 shows an example data structure of a user profile.

FIG. 10 shows an example process flow diagram implemented by a processor for carrying out monitoring and security functions described herein.

FIG. 11 shows an example process flow diagram for verifying execution of commands transmitted to user entertainment devices.

### DETAILED DESCRIPTION

In the following description of various illustrative embodiments, reference is made to the accompanying drawings, which form a part hereof, and in which is shown, by way of illustration, various embodiments in which aspects of the disclosure may be practiced. It is to be understood that other embodiments may be utilized, and structural and functional modifications may be made, without departing from the scope of the present disclosure.

FIG. 1 shows an example network **100** on which many of the various features described herein may be implemented. The network **100** may be any type of information distribution network, such as satellite, telephone, cellular, wireless, optical fiber network, coaxial cable network, and/or a hybrid fiber/coax (HFC) distribution network. Additionally, the network **100** may be a combination of networks. The network **100** may use a series of interconnected communication links **101** (e.g., coaxial cables, optical fibers, wireless, etc.) and/or some other network (e.g., the Internet, a PSTN, etc.) to connect an end-point to a local office or headend **103**. Example end-points are shown in FIG. 1 as a premises **102** (e.g., businesses, homes, consumer dwellings, etc.) The local office **103** (e.g., a data processing and/or distribution facility) may transmit information signals onto the links **101**, and each premises **102** may have a receiver used to receive and process those signals.

There may be one link **101** originating from the local office **103**, and it may be split a number of times to distribute the signal to various homes **102** in the vicinity (which may be many miles) of the local office **103**. The links **101** may include components not shown, such as splitters, filters, amplifiers, etc. to help convey the signal clearly, but in general each split introduces a bit of signal degradation. Portions of the links **101** may also be implemented with fiber-optic cable, while other portions may be implemented with coaxial cable, other links, or wireless communication paths.

The local office **103** may include a termination system (TS) **104**, such as a cable modem termination system (CMTS) in a HFC network, a DSLAM in a DSL network, a cellular base station in a cellular network, or some other computing device configured to manage communications between devices on the network of links **101** and backend devices such as servers **105-107** (which may be physical servers and/or virtual servers, for example, in a cloud environment). The TS may be as specified in a standard, such as the Data Over Cable Service Interface Specification

(DOCSIS) standard, published by Cable Television Laboratories, Inc. (a.k.a. CableLabs), or it may be a similar or modified device instead. The TS may be configured to place data on one or more downstream frequencies to be received by modems or other user devices at the various premises **102**, and to receive upstream communications from those modems on one or more upstream frequencies. The local office **103** may also include one or more network interfaces **108**, which can permit the local office **103** to communicate with various other external networks **109**. These networks **109** may include, for example, networks of Internet devices, telephone networks, cellular telephone networks, fiber optic networks, local wireless networks (e.g., WiMAX), satellite networks, and any other desired network, and the interface **108** may include the corresponding circuitry needed to communicate on the network **109**, and to other devices on the network such as a cellular telephone network and its corresponding cell phones.

As noted above, the local office **103** may include a variety of servers **105-107** that may be configured to perform various functions. The servers may be physical servers and/or virtual servers. For example, the local office **103** may include a push notification server **105**. The push notification server **105** may generate push notifications to deliver data and/or commands to the various homes **102** in the network (or more specifically, to the devices in the homes **102** that are configured to detect such notifications). The local office **103** may also include a content server **106**. The content server **106** may be one or more computing devices that are configured to provide content to users in the homes. This content may be, for example, video on demand movies, television programs, songs, text listings, etc. The content server **106** may include software to validate user identities and entitlements, locate and retrieve requested content, encrypt the content, and initiate delivery (e.g., streaming) of the content to the requesting user and/or device.

The local office **103** may also include one or more application servers **107**. An application server **107** may be a computing device configured to offer any desired service, and may run various languages and operating systems (e.g., servlets and JSP pages running on Tomcat/MySQL, OSX, BSD, Ubuntu, Redhat, HTML5, JavaScript, AJAX and COMET). For example, an application server may be responsible for collecting television program listings information and generating a data download for electronic program guide listings. Another application server may be responsible for monitoring user viewing habits and collecting that information for use in selecting advertisements. Another application server may be responsible for formatting and inserting advertisements in a video stream being transmitted to the premises **102**. Another application server may be responsible for formatting and providing data for an interactive service being transmitted to the premises **102** (e.g., chat messaging service, etc.). In some examples, an application server may implement a network controller **303**, as further described with respect to FIG. 3 below.

An example premises **102a** may include an interface **120**. The interface **120** may comprise a modem **110**, which may include transmitters and receivers used to communicate on the links **101** and with the local office **103**. The modem **110** may be, for example, a coaxial cable modem (for coaxial cable links **101**), a fiber interface node (for fiber optic links **101**), or any other desired device offering similar functionality. The interface **120** may also comprise a gateway interface device **111** or gateway. The modem **110** may be connected to, or be a part of, the gateway interface device **111**. The gateway interface device **111** may be a computing

device that communicates with the modem **110** to allow one or more other devices in the premises to communicate with the local office **103** and other devices beyond the local office. The gateway **111** may comprise a set-top box (STB), digital video recorder (DVR), computer server, or any other desired computing device. The gateway **111** may also include (not shown) local network interfaces to provide communication signals to devices in the premises, such as display devices **112** (e.g., televisions), additional STBs **113**, personal computers **114**, laptop computers **115**, wireless devices **116** (wireless laptops and netbooks, mobile phones, mobile televisions, personal digital assistants (PDA), etc.), a land-line phone **117**, and any other desired devices. Examples of the local network interfaces include Multimedia Over Coax Alliance (MoCA) interfaces, Ethernet interfaces, universal serial bus (USB) interfaces, wireless interfaces (e.g., IEEE 802.11), BLUETOOTH® interfaces (including, for example, BLUETOOTH® LE), ZIGBEE®, and others. The premises **102a** may further include one or more listening devices **119**, the operation of which will be further described below.

FIG. 2 shows an example computing device on which various elements described herein can be implemented. The computing device **200** may include one or more processors **201**, which may execute instructions of a computer program to perform any of the features described herein. The instructions may be stored in any type of computer-readable medium or memory, to configure the operation of the processor **201**. For example, instructions may be stored in a read-only memory (ROM) **202**, random access memory (RAM) **203**, removable media **204**, such as a Universal Serial Bus (USB) drive, compact disk (CD) or digital versatile disk (DVD), floppy disk drive, or any other desired electronic storage medium. Instructions may also be stored in an attached (or internal) hard drive **205**. The computing device **200** may include one or more output devices, such as a display **206** (or an external television), and may include one or more output device controllers **207**, such as a video processor. There may also be one or more user input devices **208**, such as a remote control, keyboard, mouse, touch screen, microphone, etc. The computing device **200** may also include one or more network interfaces, such as input/output circuits **209** (such as a network card) to communicate with an external network **210**. The network interface may be a wired interface, wireless interface, or a combination of the two. The interface **209** may include a modem (e.g., a cable modem), and the network **210** may include the communication links and/or networks shown in FIG. 1, or any other desired network.

The computing device **200** may include a monitoring and security application **211** that implements one or more security or monitoring features of the present description. The monitoring and security application **211** will be further described below with respect to FIGS. 3 and 4.

The FIG. 2 example is an exemplary hardware configuration. Modifications may be made to add, remove, combine, divide, etc. components as desired. Additionally, the components shown may be implemented using basic computing devices and components, and the same components (e.g., the processor **201**, storage **202**, user interface, etc.) may be used to implement any of the other computing devices and components described herein.

FIG. 3 shows an example monitoring and security system **300** for implementing features described herein. A premises may include a premises controller **301**. The premises controller **301** monitors the premises **302** and simulates the presence of a user or resident of the premises **302**. The

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premises controller **301** may monitor recorded audio signals in order to detect audio patterns of normal activities at the premises. The detected patterns may comprise, for example, indications of one or more habits of residents of the premises, for example, that a resident usually watches television in the afternoons, sometimes listens to music in the evenings, and/or other habits indicating usage patterns of media devices. When the resident is away, the premises controller **301** may command devices of the premises **302** to simulate the user's presence. For example, the premises controller **301** may turn on the television in the afternoon and turn on music in the evening to create the appearance that a resident is at home.

The premises controller **301** located in the premises **302** connects to a local office **311**, which in turn connects via the WAN **314** to the network controller **303**. The premises **302** further contains a plurality of listening devices **305** (e.g., devices that include one or more microphones) and/or video cameras **310** for monitoring the premises **302**. An alarm panel **304** connects to the premises controller **301**. Additionally, the premises controller **301** may control user entertainment devices **306**, including a television **307** and a stereo **319** via transmission(s) **316**. The premises controller **301** may also include home automation functions enabling communication with and control of lights **308** and other such devices. Various devices such as the alarm panel **304**, the listening devices **305**, lights **308**, and a video camera **310** may be connected to the premises controller **301** via a local network **312**.

The listening devices **305** may be scattered throughout the premises **302**. For example, one or more listening devices **305** may be located in each room, or in select rooms, of the premises **302**. Each listening device **305** may include one or more microphones for receiving/recording audio signals. The listening devices **305** may periodically transmit the received audio signals to the premises controller **301** for purposes of monitoring the premises **302**. The premises controller **301** may analyze and process the monitored audio signals independently or in conjunction with the network controller **303**. The listening devices **305** may send audio signals to the premises controller **301** using dedicated wires, via local network **312**, or in any other manner. A listening device **305** may be integrated with another device, such as an alarm panel **304**.

The alarm panel **304** may control security settings of the monitoring and security system **300**. For example, a user may change an arming mode of the monitoring and security system **300** via the alarm panel **304** in order to enable or disable certain security features. Arming modes may include an "away" mode, a "night" mode, and/or a "stay" mode, among others. The premises controller **301** may check the modes set at the alarm panel **304** in order to determine a mode of the premises controller **301**. When a mode indicates a user is present, the premises controller **301** may monitor the premises **302** to detect patterns of normal activity and behavior. When a mode indicates a user is away, the premises controller **301** may simulate the user's presence at the premises.

A portable communication device **317** (e.g., a smartphone) and/or a personal computer **318** may connect to the premises **302** via the WAN **313** (in conjunction with cellular network **315**) and/or the WAN **314**. The portable communication device **317** and/or the personal computer **318** may communicate with the network controller **303**, which may in turn relay communications to and from the premises controller **301**. Such communications may include requesting information from the security system, modifying a setting,

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or the like. For example, a resident could modify a user profile generated by the premises controller **301** in order to determine what actions the premises controller **301** takes in the user's absence from the premises **302**.

Additionally or alternatively, the portable communication device **317** and/or the personal computer **318** may communicate with the premises controller **301** without the involvement of the network controller **303**. The network controller **303** may perform the functions ascribed herein to the premises controller **301** instead of or in addition to the premises controller **301**. The network controller **303** may be integrated with the local office **311** (e.g., as an application server **107** as shown by FIG. 1). Accordingly, an application server **107** embodying the network controller **303** may perform any of the techniques described herein.

The premises controller **301** may be implemented as a hardware or software component of the computing device **200** (e.g., as a monitoring and security application **211**). Additionally or alternatively, the premises controller **301** may be implemented as a standalone device.

FIG. 4A shows the monitoring and security system operating in a first mode in which a user **320** is at the premises **302** (e.g., a stay mode). The user **320** may interact with one or more user entertainment devices **306**, such as the television **307** and the stereo **319** (e.g., using remote controls or other user interfaces). The user entertainment devices **306** may generate sound. Additionally, the user **320** may generate sound, for example, by speaking. The listening devices **305** may record the sounds generated in the premises **302** and send the recorded audio signals to the premises controller **301** for analysis.

The premises controller **301** may analyze each audio signal to determine or estimate meta information about the audio contained in the signal, such as a volume of the audio signal, a classification of the audio signal (e.g., speech, music, etc.), and/or a device or person which may have generated the audio signal. The recorded audio signal together with the meta information may be stored as an event in an event log. The premises controller **301** may further analyze the event log to determine information about patterns of activity.

The premises controller **301** may analyze several days' worth of logged events in order to determine a typical day for a user **320**. After finding patterns characterizing the typical day, the premises controller **301** may create a schedule of commands that may be sent to the user entertainment devices **306** to simulate the presence of the user **320**.

Additionally, the premises controller **301** may analyze the event logs to determine pairs of events. For example, the premises controller **301** may determine that a listening device **305** frequently records a dog barking after a doorbell rings. The premises controller **301** may create a dynamic event that can be triggered by the first event of the pair. Accordingly, when a user is away and the premises controller **301** detects a doorbell ringing, it may instruct a stereo to play the sound of a dog barking. Accordingly, a user profile may be updated with dynamic events that further characterize patterns of activity for the premises **302**.

FIG. 4B shows the monitoring and security system operating in a second mode in which a user **320** is away from the premises **302** (e.g., an away mode). The premises controller **301** may send commands to the user entertainment device **306** at certain times as instructed by the schedule of commands. Additionally, the listening devices **305** may continue monitoring the premises for dynamic event triggers. When



such a trigger event is detected, an appropriate command may be sent to a user entertainment device **306** in response to the dynamic event.

FIG. 5 shows an example hardware embodiment of the premises controller **301**. The premises controller **301** contains a processor **501** comprising several modules **502-507** for implementing functions described herein. The processor **501** stores data and accesses data at the storage **508**. The processor **501** also transmits (and/or receives) transmissions from transceivers **510a-n**. The processor **501** connects to a local network interface **509**, which may further connect (via a local network **312**) to a gateway **111**. Signals **511** may be received at the processor from monitoring devices (e.g., listening devices **305** and/or video cameras **310**, etc.).

The processor **501** includes a command generator **502** that generates commands or sequences of commands for transmission to user entertainment devices **306** via transceivers **510a-n**. The command generator **502** may verify that the commands were successfully executed by listening to audio signals received by listening devices **305**. For example, after sending a command to a television via an infrared signal, the command generator **502** may check signals **511** to ensure that appropriate sounds are being generated by the television. Accordingly, the command generator **502** may be configured to execute sub-routines to verify that a command or commands were successfully executed by the target user entertainment device **306**. The audio analyzer **503** analyzes audio signals received from listening devices **305** in order to detect and/or log audio events. The audio signals may capture sounds generated by user entertainment devices **306**, users or residents of the premises **302** (e.g., speech), other devices (e.g., appliances) of the premises **302**, or any other sources of audio at the premises **302**. The audio analyzer **503** may analyze the audio signals in order to determine meta information about the audio signals. The audio analyzer **503** may determine or estimate a volume of the audio signal, a classification of the audio signal, a room of the audio signal, a device or person that generated the audio signal, as well as other properties.

Similarly, the video analyzer **505** may analyze video signals received from the video cameras **310** in order to detect and log video events. For example, the video analyzer **505** may analyze video signals to determine when lights are on or off. The signals **511** may be received from dedicated interface(s) of the premises controller **301**, via one or more of transceivers **510a-n**, and/or via the local network interface **509**. Signals **511** may be received periodically or continuously from the listening devices **305** and/or video cameras **310**.

The profile Generator **504** may generate one or more user profiles **512** indicating typical behavior for one or more users. A user profile **512** may be generated based on a log **515** of events analyzed by the audio analyzer **503**, video analyzer **505**, and/or automation controller **507**. For example, if the user typically watches a news program from 7 am to 8 am, listens to music on a stereo from 1 pm to 3 pm, and has the office lights turned on from 7 am to 9 pm, then this behavior may be detected by the audio analyzer **503** and video analyzer **505**, and information may be stored in the user profile to indicate this behavior. The user profile **512** may also include a schedule of commands that can be sent to user entertainment devices and/or other controlled devices to create the appearance that one or more residents is at home by, for example, indicating a predetermined time (and/or a randomized time with respect to a predetermined time) when a video device should be activated, and a setting or service that should be activated (e.g., turning on a

television and tuning to the news program from 7 am to 8 am, tuning to a stereo should from 1 pm to 3 pm, and turning on the office lights from 7 am to 9 pm). When the system determines that the house is unattended, the system may send automatic commands to control the television/stereo/lights to mimic the same behavior.

The security monitor **506** may implement security functions such as setting and triggering alarms, detecting break-ins, etc. The security monitor **506** may further send alerts or messages to remote devices such as the portable communication device **317** and/or personal computer **318**. The automation controller **507** may implement home automation protocols for controlling lights and other smart devices via one or more home automation protocols (e.g., X10, Z-WAVE®, INSTEON®, ZIGBEE®, and other such protocols). The automation controller **507** may also monitor signals transmitted to and/or received from smart devices indicating events (e.g., lights being turning on or off) and log such events in the log **515**.

Each of the modules **502-507** may be implemented as software modules, components of software programs, combinations of hardware and/or software components, or the like.

All or a portion of the monitoring and security application **211** may be implemented by the processor **501** and any or all of the modules **502-507**. Any or all of the modules **502-507** may be implemented at the network controller **303** instead of or in addition to at the premises controller **301**.

Storage **508** includes one or more user profiles **512** generated by the profile generator **504**, one or more device profiles **513** specifying controlled devices and their capabilities, one or more source maps **514** for mapping media content to particular media sources, a log **515** of events detected from monitoring device signals **511**, media samples **516** for detecting and playing back events, and other data needed by the premises controller **301**. Storage **508** may include any type of storage device(s), such as hard drives, flash memory, ROM, RAM, and the like.

The device profiles **513** may indicate device-specific information such as what protocol a user entertainment device **306** uses to receive commands, how commands should be formatted, and the like. The device profile **513** may further specify a room where the device is located. The audio analyzer **503** may determine or estimate, based on information indicating the room, which device (or devices) may have generated a sound recorded by a listening device **305**. Additionally, the premises controller **301** may select, based on information indicating the room, which user entertainment device **306** it should command. For example, if a user profile indicates that a resident frequently talks on the phone in the living room, the premises controller **301** may simulate that behavior in the resident's absence by playing an audio file containing speech on the living room stereo. In this example, the device profile **513** for the stereo indicates it is in the living room.

Source maps **514** may indicate sources of content (e.g., television channels) corresponding to remote control commands for a particular service. For example, when a premises controller **301** executes a command to tune a television to MSNBC, it may access the source map **514** to determine the appropriate channel number to send to the television or set-top box. The source map **514** may further indicate which sources are free to access. In some examples, the premises controller **301** selects sources from the subset of sources that are free to access in order to avoid accruing charges for simulating a user's presence in the premises **302**.

Media samples **516** may include media files pre-installed in the storage **508**, uploaded by a user, and/or downloaded from the network. The media samples **516** may be used by the audio analyzer **503** to recognize certain sounds. For example, the user may upload a recording of the doorbell so that the audio analyzer **503** can recognize the user's doorbell and classify a corresponding event appropriately. Accordingly, the audio analyzer **503** can compare media samples **516** to audio signals received from listening devices **305** to determine metadata about the audio signals. In this example, the media sample **516** may be tagged (e.g., by the user **320**) with metadata such as a classification of "doorbell," a particular room, and/or other such metadata.

Additionally, the media samples **516** may be used by the premises controller **301** to play certain sounds. For example, a recording of a dog barking could be played on the stereo to simulate the presence of a dog when a doorbell event is detected. Accordingly, the premises controller **301** may download a media sample **516** associated with the "dog barking" classification when a profile specifies such a dynamic event. Additionally or alternatively, the user **320** may manually upload such a media sample **516**, or such a media sample **516** may come pre-installed in the premises controller **301**.

Transceivers **510a-n** may include one or more transmitters and/or receivers for communicating with user entertainment devices **306** and other devices such as lights **308**. For example, a first transceiver **510a** may implement a BLUETOOTH® protocol. A transmitter **510b** may be an infrared (IR) blaster for emulating remote control commands, such as commands for a television **307** or stereo **319**. A transceiver **510c** may implement a radio frequency communication protocol, such as ZIGBEE® or WI-FI®. Other transceivers **510n** could be used as well.

FIG. 6A shows example data structures including example logs **601a-b**, groups **602a-c** of events, and habits **603a-b**. Logs **601** may include a plurality of events including meta information about each event. Such meta information may include, as shown in FIG. 6A, time information, classification information, volume information, and room information. The meta information may be determined by the audio analyzer **503** and stored in the log **601** for analysis.

The profile generator **504** may analyze the logs **601** and generate data for constructing a user profile. The profile generator **504** may perform a two-step process for generating groups **602** of events from logs **601**, followed by generating habits **603** from groups **602** of events. Additionally or alternatively, the profile generator **504** may use other processes to analyze the log **601** data and generate a profile therefrom.

FIG. 6A shows a portion of a log **601a** for a first day and a portion of a log **601b** for a second day. The profile generator **504** may group events by common values of meta information. The profile generator **504** may group events by time and room information in order to generate groups **602** of events. For example, the profile generator **504** may determine that multiple events in a first log **601a** occur in a time period of 1:10-1:14 in the living room. The profile generator **504** may generate a group **602a** corresponding to those events.

Such groups **602** may be formed of events having relatively contiguous time values for a given room. For example, if a log **601a** contains another event entry at time 1:46 in the living room, the profile generator **504** may avoid adding such an event to the first group **602a** because 1:46 is too far in time from the remaining members of the group.

Events over a threshold amount of time (e.g., 30 minutes) difference from other events may be split into different groups **602**.

In the example of FIG. 6A, the profile generator **504** may analyze the events corresponding to the group **602a** and determine that the majority of such events have a "speech" classification. Therefore, the profile generator **504** may assign a "speech" class to the group **602a**. The profile generator **504** may detect that above a threshold percentage of events in a group **602** have a single class value, and assign that class value to the group **602**. For example, if more than 75% of events in a group **602** have a "speech" value, the entire group may be assigned a "speech" value.

The profile generator **504** may further determine that all of the events in the example group **602a** (or a subset of events, for example the subset of events classified as "speech") have volume properties within a given range (e.g., 60-65 dB). Accordingly, the profile generator **504** may assign the given volume range to the group **602a** as shown in FIG. 6A.

The profile generator **504** may determine multiple groups **602** from a single log **601a**. For example, the profile generator **504** may determine a second group **602b** corresponding to events occurring between 1:10-1:16 in the basement. The profile generator **504** may assign a classification and a volume range to the group **602b** in a similar manner as for the group **602a**.

The profile generator **504** may determine additional groups from a log **601b** for a second day. The profile generator **504** may be configured to assign multiple classes to a single group **602**, such as when events of the group **602** are split fairly evenly between classifications. For example, because the log **601b** contains a number of events with both music and speech classifications, the group **602c** may be assigned two classifications as shown in FIG. 6A. The remaining properties of the group **602c** may be assigned as for the other groups **602a**, **602b**.

The profile generator **504** may form groups **602** based on one or more other properties instead of or in addition to time and room properties. For example, the profile generator **504** may form groups **602** of events having the same classification, such that each group has at most one classification. Additionally, the profile generator **504** may form groups **602** based on how close in volume events are to each other, such that relatively loud events may be split into different groups from relatively quiet events. Additionally, any other properties included as meta information for an event may be used for forming groups **602**. Such properties may include a device or person that generated the sound, a type of the event, a frequency property of the event, or any other property of the event.

After forming the groups **602** of events, profile generator **504** may further analyze the groups **602** to form habits **603**. Habits **603** may be formed by comparing groups **602** across different days. Groups **602** corresponding to two or more days may be compared. When profile generator finds matching (or partially matching) groups of events across multiple days, it may create a habit **603**. For example, when profile generator **504** determines that groups **602a** and **602c** are partial matches in terms of class (e.g., both groups contain a "speech" class), time of day (e.g., the groups time ranges are relatively close together), and room (e.g., both groups indicate the living room), it may generate a habit **603a** based on the properties of groups **602a** and **602c**.

The profile generator **504** may determine a partial match based on one or more values of meta information being the same or similar. In some examples, the profile generator **504**

may generate a habit if the time information is similar (e.g., if time ranges for the two groups overlap at least partially, or have less than a certain amount of time between an end time of one group and a start time of the other, or if some other measure of time similarity indicates the time information is similar). In the example of FIG. 6A, the profile generator 504 may determine that groups 602a and 602c are similar in terms of time because they occur within a threshold time of each other (e.g., 2 hours). Other thresholds may be used to determine similarity of other properties (e.g., within 10 dB for volume).

The profile generator 504 may determine that two groups are a partial match even when some properties are not the same or similar. For example, two groups 602b and 602c may be determined as partial matches even though the room values are different.

The profile generator 504 may generate the habits 603 based on the properties of the matched groups 602. For example, the time range of a habit 603a may be set using the earliest and latest times of corresponding groups 602a and 602c, the volume range may be set using the quietest and loudest volumes of corresponding groups 602a and 602c, etc. The profile generator 504 may determine the properties of a habit 603 based on the union/overlap of the property values of the matched groups, the average values of the properties of the matched groups, or based on some other method of combining the properties of two or more groups. For example, the value of the room property for a habit 603b may include the values of both room properties for matching groups 602b and 602c.

The profile generator 504 may use the habits 603 in order to generate commands for simulating a user's presence. Such commands may be generated taking into account the devices connected to the system and a user's preferences. The generated commands may be added to a schedule of the user profile 512.

FIG. 6B further shows a pair 604 of events detected from multiple days' logs 601a, 601b. In the example of FIG. 6B, the profile generator 504 detects that events classified as "doorbell" and "dog" frequently occur together in a particular order. Accordingly, the profile generator 504 may detect that every time (or above a threshold percentage of the time) a first event occurs, a second event occurs. The profile generator 504 may create a pair 604 including the first and second events. The pair 604 may be used to create dynamic events so that a premises controller 301 can dynamically simulate a user's presence.

FIG. 7A shows an example user interface 700 by which a user 320 can adjust a schedule of commands generated by the profile generator 504 based on the habits 603. The profile generator 504 may generate commands based on properties assigned to the habits 603. For example, commands for turning the television on and off may be generated based on a habit 603a.

The profile generator 504 may select a device 702 to receive the one or more commands based on properties of the habit 603. Command preferences may specify mappings between properties of a habit 603 and a device 702 that will receive a command. For example, command preferences may map a living room television to a class property indicating "speech" for the habit 603a. The command preferences may indicate that "music" habits map to a stereo device. The command preferences may further indicate that other properties (e.g., a property indicating a room) map to particular devices 702.

The profile generator 504 may choose the selected device 702 from among several devices indicated by the command

preferences based on a room property. For example, the profile generator 504 may select the living room television as indicated by the room property of habit 603a.

After choosing a device 702, the profile generator 504 may construct one or more commands for the device 702 based on additional properties of the habit 603, one or more device profiles 513, a source map 514, and additional command preferences. For example, the profile generator 504 may select "turn on" and "turn off" actions 701 based on the device profile 513 for the living room television indicating it accepts those commands. In this example, the profile generator 504 may select "NBC" for an optional source parameter 703 based on a source map 514 and command preferences. For example, command preferences may indicate that NBC should be selected to emulate a speech habit. The profile generator 504 may further select times 704 for the commands based on time properties of a habit 603.

Similarly, the profile generator 504 may select a stereo device 702 that matches a class property indicating "music" for a habit 603b. The remaining actions 701, optional parameters 703, and time 704 for the corresponding commands may be generated by profile generator 504 as discussed above. Particularly, actions 701 may be determined based in part on capabilities of a device that will receive the command. Such capabilities may include particular commands or types of commands that a device can execute. The profile generator 504 may retrieve the capabilities of a device from a corresponding device profile 513. Optional parameters 703 may be determined based on command preferences indicating, for example, one or more preferred sources of media content. Time 704 may be determined based on a time property of the habit 603 (which may reflect, for example, one or more times associated with a plurality of logged events, or one or more times associated with patterns of events).

The user interface 700 may be an interactive user interface by which a user 320 can edit the scheduled commands. For example, a user 320 may select and/or interact with an element 701-704 of one of the commands in order to adjust it. Therefore, a user 320 may modify the schedule of commands as desired.

FIG. 7B depicts a user interface 705 by which a user 320 may view and modify a list of dynamic events generated by the profile generator 504. The dynamic events may be generated based on a pair 604 of events detected by the profile generator 504. The profile generator 504 may select the first event of the pair 604 (e.g., a doorbell ringing) as the trigger and create a command that will be executed when the first event is detected (e.g., playback of a sound of a dog barking).

As before, the profile generator 504 may select a device 702 to receive one or more commands based on properties of a pair 604 of events. For example, to emulate a dog barking, the profile generator 504 may play a corresponding sound file on the stereo. The commands may be configured based on command preferences, device profiles 513, and any other settings.

The user 320 may interact with the interface 705 to edit the list of dynamic events. The user may add a dynamic event by selecting a trigger 707, optionally uploading 708 a sound file corresponding to the trigger, and/or setting one or more of actions 701, device 702, and parameters 703 for the command. The profile generator 504 may create a dynamic event that triggers when the selected trigger 707 is detected (which may be based on detecting a match with the sound

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file 708). When the trigger 707 is detected, the corresponding command may be executed.

Users 320 may access the user interfaces 700, 705 via a user device, such as a portable communication device 317 or personal computer 318. The user interfaces 700, 705 may be generated by the premises controller 301 and/or network controller 303.

FIG. 8 shows a user interface 800 by which a user 320 may identify an event that the audio analyzer 503 was unable to classify. When the audio analyzer 503 is unable to classify an event, it may send an alert to a user device, such as the portable communication device 317 or personal computer 318. A user of the user device may access the user interface 800 to resolve the unidentified event.

The user interface 800 may include a button 801 for playing back the unidentified event. The premises controller 301 may store a recording of the unidentified event in storage 508 so that a user can play back the recording upon selection of the user interface playback button 801.

The user interface 800 may further include a selector 802 for selecting a classification of the unidentified event after the user listens to the recording. For example, the audio analyzer 503 may have indicated one or more classifications with low confidence. Accordingly, the selector 802 may contain the indicated classifications indicated along with confidence levels determined by the audio analyzer 503. The classifications may be ranked in order of their determined confidence levels. Additionally or alternatively, the selector 802 may present a list of all classifications used by the system for selection by the user.

The selector 802 may further include an additional “not sure” option for indicating that the user cannot identify the sound. The selector 802 may additionally include an “ignore” option that causes the sound (and matching sounds) to be ignored by audio analyzer 503 (e.g., not logged and/or not simulated) in the future.

The user interface 800 may include additional selectors for indicating other properties. For example, a selector 803 may be used to indicate a device from which the sound originated. The list of devices in the selector 803 may be limited to devices in the same room as the listening device 305 that recorded the sound.

After a user indicates the correct values for the event, a log 601 may be updated with the resolved event data. The profile generator 504 may re-analyze the updated log 601 after unidentified events have been resolved by the user.

FIG. 9 shows a data structure for the user profile 512. The user profile 512 may comprise a command schedule 901 for storing the schedule of commands generated by the profile generator 504, dynamic events 902 for storing the dynamic events generated by profile generator 504, privacy options 903 for controlling listening devices 305, and command preferences 904 used to generate the schedule of commands and dynamic events.

The command schedule 901 and dynamic events 902 may store commands for operating user entertainment devices 306. The command schedule 901 and dynamic events 902 may also store additional data about commands, such as which transceivers 510 to use to send a particular command, the format and/or protocol of commands, and one or more scripts for verifying that a command was successfully executed. The profile generator 504 may select a transceiver 510 based on the capabilities of a device 702 of the command. The profile generator 504 may further store a format and/or protocol for the command as indicated by the device profile 513 for the device 702. For example, a television 307 may be configured to accept commands via an IR blaster

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510b. Accordingly, the profile generator 504 may store (e.g., along with a command for the television 307) an indication of what data to send via the IR blaster 510b in order to execute a command. Additionally, one or more scripts may be provided that verify a command was executed. An example script is described below with respect to FIG. 11.

Privacy options 903 may indicate times and/or modes for which selected (or all) listening devices 305 should be disabled. For example, privacy options may indicate that all listening devices 305 should be disabled in a night mode, or between certain hours, or that only particular listening devices 305 should be disabled in certain modes or times. The audio analyzer 503 may check the privacy options 903 in order to ignore audio signals received from particular listening devices 305 at time and/or modes specified by the privacy options 903.

Command preferences 904 may indicate preferred settings for constructing commands by profile generator 504. For example, a user may prefer certain television or audio sources (which may be specified as parameters 703) for emulating habits 603, certain devices 702 for emulating habits 603, and the like. Accordingly, the command preferences 904 may contain mappings that indicate how certain properties of habits 603 indicate parameters 701-703 for commands. The command preferences 904 may additionally or alternatively contain default mappings that a user 320 can override. For example, a default mapping may indicate that a news channel may be tuned on a television to emulate a habit with a “speech” class. However, a user may override the default preference and indicate, for example, that a sports channel should be tuned on the television to emulate the habit with a “speech” class.

FIG. 10 shows an example process flow for processor 501 of premises controller 301. The premises controller 301 may be turned on at step 1001. Before entering a main loop, the premises controller 301 may execute configuration steps 1002, 1003. At step 1002, the premises controller 301 may configure one or more user profiles 512. On a first boot, the premises controller 301 may create a default user profile 512 and/or generate configuration prompts for creating one or more user profiles 512 based on inputs from user(s) 320. A user may be prompted for favorite sources that may be stored in command preferences 904 of the user profile 512. A user may be presented with a user interface 700 for configuring a default command schedule 901 of the user profile 512.

At step 1003, the premises controller 301 may configure one or more devices, such as listening devices 305, user entertainment devices 306, or other premises devices. The premises controller 301 may establish communications with such devices in order to determine device capabilities and populate a device profile 513 for each device. Additionally or alternatively, the premises controller 301 may prompt the user 320 for information about devices in order to configure device profiles 513. Steps 1002 and/or 1003 may be executed at any time based on receiving a communication from a user device, such as a portable communications device 317 or personal computer 318. For example, a user 320 may configure a user profile 512 at any time by connecting to the premises controller 301 via the Internet.

At step 1003, the premises controller 301 may determine the mode of the security system. The premises controller 301 may regularly poll the alarm panel 304 of the security system to determine the mode, or the alarm panel 304 may send a message to the premises controller 301 when the mode is changed automatically or by a user 320. When the

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mode is set to “stay,” the premises controller **301** may proceed to decision **1005** to determine whether an event is detected.

At decision **1005**, when the mode is set to “stay,” the premises controller **301** may monitor events to determine whether an event has been detected from audio received by listening devices **305** and/or video received from a video camera **310**. The audio analyzer **503** of premises controller **301** may continually receive and process audio signals from listening devices **305** to detect events. For example, the audio analyzer **503** may store the most recent N seconds or minutes of audio from each listening device **305** (e.g., in a circular buffer). The audio analyzer **503** may periodically or continually receive and analyze portions of any or all of the audio signals in order to detect events. The audio analyzer **503** may measure temporal characteristics of the audio signals, such as volume over time or change in volume over time, in order to detect an event. Thus loud noises recorded in the audio signals may be flagged as events. Accordingly, the audio analyzer **503** may indicate an event when the volume of one or more audio signals is above a threshold (e.g., 55 dB), remains above a threshold for a certain duration of time, and/or changes in volume at certain rates.

The audio analyzer **503** may additionally or alternatively analyze frequency characteristics of the audio signal, such as frequency components, spectral centroid, spectral flux, and/or spectral density to detect events. The audio analyzer **503** may flag signals containing certain frequency characteristics as events. The audio analyzer **503** may use Fourier transforms, wavelet transforms, and the like to obtain the frequency characteristics of the signal to detect an event.

Upon detecting an event, the audio analyzer **503** may store an indication of an event along with the audio corresponding to the detected event. Accordingly, when an event is detected at decision **1005**, the premises controller may analyze the event to determine its properties at step **1006**.

The premises controller **301**, via the audio analyzer **503**, analyzes events in order to determine information about the event that can be stored in a log **601** for later analysis by the profile generator **504**. The audio analyzer **503** may analyze features of the audio signal in order to determine properties of the event. The audio analyzer **503** may use one or more classification algorithms to analyze the audio signals corresponding to each event. Accordingly, the audio analyzer **503** may extract one or more features from the audio signal and classify the detected event using one or more classification algorithms. Such features may include time-domain features (such as short-time energy, average zero crossing rate, etc.), frequency-domain features (such as mel frequency cepstral coefficients, spectral flux, spectral centroid, etc.), or other features of the audio signal. The classification algorithms may determine whether a detected event is recognized by a classifier trained to categorize audio signals as matching one or more of a defined set of classes (e.g., speech, music, a dog barking, etc.). Various classification algorithms, such as linear classifiers, support vector machines, neural networks, and other such classification algorithms can be used by the audio analyzer **503** to analyze the events at step **1006**.

The audio analyzer **503** may additionally detect a volume property of an event and assign a corresponding volume property. The audio analyzer **503** may calculate an average volume of the event, a max volume of the event, or some other measurement of volume. The audio analyzer **503** may optionally use a frequency weighting filter such as A-weighting, or ITU-R **468** noise weighting, in order to

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more realistically measure perceived volume. Thus audio analyzer **503** may further assign a volume property as part of step **1006**.

The audio analyzer **503** may additionally determine or estimate a room associated with the event and assign a corresponding room property. The audio analyzer **503** may determine or estimate the room based on which listening device **305** recorded the audio signal corresponding to the event. The audio analyzer **503** may look up the room associated with the listening device **305** that recorded the corresponding audio signal (e.g., from a device profile **513** for the listening device **305**), and assign the room property accordingly. Multiple listening devices **305** may generate similar events at the same time (e.g., because multiple listening devices recorded the same event). The audio analyzer **503** may detect that similar audio signals were recorded by multiple listening devices **305** by comparing the audio signals, or the properties of events detected from the audio signals. Accordingly, the audio analyzer **503** can estimate a room or rooms where the event may have occurred based on the volume and/or timing of each received audio signal (e.g., the highest volume and/or earliest received audio signal may indicate the closest listening device **305**). Additionally, the audio analyzer **503** may use relative timing to compare the audio signals to “triangulate” a distance and/or location of the audio event relative to the listening devices **305** in order to determine a room. Each listening device **305** may contain multiple microphones (e.g., an array of directional microphones) generating multiple audio signals in order to assist in room estimation. Such listening devices **305** could process the multiple audio signals locally in order to estimate a direction and/or location of the audio event before transmitting that information to the audio analyzer **503**. Accordingly, the audio analyzer **503** may assign room properties as part of step **1006**.

Other properties may be assigned by the audio analyzer **503**. For example, the one or more classifiers may further indicate sub-classifications, such as a genre of a music class, a gender or age for a speech class and the like. The audio analyzer **503** may identify a particular piece of media content, for example, by comparing a fingerprint of the audio signal to a database of media content fingerprints. Such a comparison could be used to identify additional meta information about the audio signal. The audio analyzer **503** may also estimate a device that generated an event based on the room and class properties. For example, if an event is classified as “music” and has a room property indication “living room,” the audio analyzer **503** may estimate that a living room stereo generated the event and assign a corresponding device property.

At step **1007**, the premises controller **301** stores the analyzed event (e.g., indicators of the event and its associated properties determined by the audio analyzer **503**) in a log **601**. The log **601** may additionally contain a timestamp marking the time when an event was recorded by a listening device **305**. Thus, the premises controller **301** may build up the log **601** over time so that it can be processed by the profile generator **504**.

The audio analyzer **503** may attempt to classify an audio signal corresponding to an event, but classification algorithms may indicate a low confidence of an accurate classification. In that case, the event may be classified as “unidentified” in the log **601**. At decision **1008**, the premises controller **301** may check whether the log **601** contains an event classified as “unidentified.” When such an event is detected, the premises controller **301** may send an alert to a user device so the user **320** can identify the event if desired.

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The alert may contain a link by which a user can access a user interface **800** for resolving unidentified events. The user identification of the event can be used to re-train or otherwise improve classifiers.

At decision **1010**, the premises controller **301** may determine whether to update a user profile. The premises controller **301** may cause the profile generator **504** to update the user profile **512** at regular intervals, e.g., daily or weekly. Additionally or alternatively, the profile generator **504** may update the user profile **512** when a log **601** contains above a threshold amount of data, above a threshold number of events, or some other measure indicating an amount of data in the log **601**. The profile generator **504** may update the user profile **512** regardless of the alarm mode. When the profile generator **504** determines to update the user profile **512**, it processes data in the log **601** to determine a command schedule **901** and/or dynamic events **902**.

At step **1011**, the profile generator **504** creates groups **602** of events having common properties (e.g., events that occur in common rooms within common time frames). The profile generator **504** thus summarizes what was happening within the premises **304** at given times. The groups **602** of events may be assigned a particular class common to the events within the group **602** (or above a threshold number or percentage of the events within the group **602**). For example, a group may be assigned a “speech” class because a majority of the events within the group were classified as speech events.

The groups **602** of events may be assigned a volume property in addition to or as an alternative to the class property. The assigned volume property may be an average of the volume properties assigned to the events of the group **602**, or it may be a range encompassing all (or a majority) of the volume properties of the events of the group **602**.

At step **1012**, the profile generator **504** may compare two or more event groups **602** across multiple days in order to determine patterns of activity at the premises. The profile generator **504** generates habits **603** that indicate such patterns of activity based on the group comparison (e.g., by combining properties of the two or more groups **602**, averaging such properties, finding the union of such properties, etc.).

At step **1013**, the profile generator **504** may map the habits **603** to one or more commands that emulate the habits **603** when a user is away. For example, a habit **603** indicating a user often listens to music in the afternoon may map to a command to turn on a user’s stereo in the early afternoon and turn it off in the late afternoon. The mappings of habits to commands may be based on one or more command preferences **904**. The command preferences **904** may indicate what types of commands should be generated based on properties of habits **603**. For example, a habit with a “speech” property may be mapped to a particular device (e.g., a television) and source (e.g., MSNBC), so that the profile generator **504** creates appropriate commands to operate the device at a time indicated by the habit **603**.

At step **1014**, the profile generator **504** searches for pairs **604** of events in the logs **601**. Pairs **604** may include one or more events that frequently occur together in a particular order. Examples of such pairs **604** may be “phone ringing” followed by “speech,” “doorbell ringing” followed by “dog barking,” “music” followed by “dim lights,” and the like. The profile generator **504** may find the pairs **604** in the log **601** by determining that above a threshold percentage of the time a first event is detected, a second event is detected within a particular time frame (e.g., 30 seconds). Additionally or alternatively, the profile generator **504** may find such

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pairs **604** based on determining that two events occur in a particular order (e.g., in a particular time frame) above a threshold number of times.

At step **1015**, the profile generator **504** generates dynamic events **902** based on the pairs of events. The first event of the pair may be set as a trigger **706** for sending a command when an event corresponding to the trigger **706** is detected. For example, a command may be generated to play an audio recording of speech after premises controller **301** detects a phone ringing event. The profile generator **504** may use command preferences **904** in a similar fashion as for step **1013**.

At step **1016**, the profile generator **504** updates the command schedule **901** of the user profile **512** with the commands generated at step **1013**, and updates the dynamic events **902** of the user profile **512** with the triggers and commands generated at step **1015**. The update of the profile **512** may then be completed and the process flow may return to decision **1004** and restart the “stay” mode loop.

When the mode is set to “away,” the premises controller **301** attempts to emulate patterns of behavior at the residence using the command schedule **901** and dynamic events **902** of the user profile **512**. Accordingly, when the premises controller **301** detects an away mode at decision **1004**, the premises controller **301** may proceed to decision **1020**. At decision **1020**, the premises controller **301** may compare the current time to the command schedule to determine whether any scheduled commands are due for execution.

When a current time matches a time in the command schedule, at step **1022** the premises controller **301** causes the command generator **502** to transmit a corresponding command (or commands) to one or more user entertainment devices or other devices in accordance with the command schedule **901**. A single entry in the command schedule may contain multiple commands for one or more devices. For example, a scheduled entry for tuning a television to a certain channel may include multiple commands for turning on a television, changing the volume of the television, changing the channel of a set top box, etc. In addition to or as an alternative to sending a command at an exact time indicated by the command schedule **901**, the command generator **502** may transmit the command at a randomized time with respect to the time indicated in the command schedule **901**. For example, a command scheduled for 7:00 PM may be transmitted at a random time within 15 minutes before and/or after 7:00 PM.

At step **1023**, the command generator **502** may optionally verify that the command (or commands) was successfully executed. The command generator **502** may execute one or more sub-routines to verify that an action was successfully performed by the instructed user entertainment device based on feedback from listening devices **305**. For example, the listening devices **305** may record an audio signal containing music after the command generator **502** sends a command to play music, which may indicate that the command was successfully executed. An example of such a sub-routine is further described with respect to FIG. 11.

At decision **1021**, the premises controller **301** monitors listening devices **305** (e.g., via audio analyzer **503**) to detect an event that matches a dynamic event trigger **706**. The audio analyzer **503** may detect events in a similar manner as for decision **1005** and analyze the event in a similar manner as for step **1006**. The audio analyzer **503** may use the same classifiers as for step **1006** to determine properties of the event before matching the event to a dynamic event trigger **706**. The audio analyzer **503** may use different classifiers in addition to or instead of the classifiers described with respect

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to step **1006**. For example, the audio analyzer **503** may use real-time classifiers to detect and/or analyze events at decision **1021**. Real-time classifiers may be optimized to quickly detect short-term events such as a “doorbell” or a “phone ringing.”

When a dynamic event trigger **706** is detected, at step **1022** the command generator **502** sends a command specified by the dynamic events **902** in a similar way as for the command specified by command schedule **901**.

The process flow shown in FIG. **10** and described above is an example operational representation of the process executed by the processor **501** and is not meant to imply any particular programming techniques, hardware structure, or code structure. For example, programming techniques that do not use mode-switched loops and/or event-driven architecture are meant to fall within the ambit of the disclosure.

FIG. **11** shows an example process for verifying execution of a command sent to a user entertainment device **306** by the command generator **502**. Such a process may be used, for example, by the command generator **502** in order to tune a television **307** to a particular source (e.g., the “Turn on TV to NBC” example of FIG. **6**). The process of FIG. **11** is exemplary, and other processes could be used by the command generator in order to carry out other types of commands.

Prior to block **1101**, the command generator **502** generates and transmits (e.g., via IR transmitter **510b**) a command to set the user entertainment device to a selected media content. For example, the command generator **502** may formulate one or more IR commands to effect a media source change (e.g., a channel change) or a media input change using a format recognized by a television **307**. The command generator **502** may access a device profile **513** that specifies how commands for a particular device should be formatted. Additionally or alternatively, the command generator **502** may access source maps **514** that indicate what sources correspond to particular content.

After sending the command, the processor **501** and/or command generator **502** may be unaware of whether the command was successfully executed by the instructed user entertainment device **306**. Many user entertainment devices **306** use one-way communication links such as infrared. These devices **306** often lack a back channel to indicate what state the device **306** is in so that successful execution of a command can be verified. For example, an infrared command sent to a television **307** might fail because the television is off, the television is set to the wrong input, or something blocked the line of sight between the IR transmitter **510b** and the television **307**.

At block **1101**, the command generator **502** may instruct the audio analyzer **503** to analyze an audio signal received after the command was sent in order to determine whether the command was successfully executed. For example, the command generator **502** may instruct the audio analyzer **503** to analyze an audio signal from a listening device **305** in the same room as the instructed user entertainment device **306**. The audio analyzer **503** may analyze the signal as described at step **1006** in order to determine properties of the audio signal.

At decision **1102**, based on the analysis of the audio signal at block **1101**, the command generator **502** determines whether a class of the audio signal matches the transmitted command. For example, a class property of “music” or “speech” may indicate that a command to turn on a television was successfully executed (e.g., because sounds from a television often contain music or speech). The command generator **502** may use command preferences **904**, which

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may map classes to particular commands, in order to detect which classes indicate successful execution of a particular command or commands.

Additionally or alternatively, at decision **1102** the command generator **502** may detect a match of one or more other properties in addition to or as an alternative to a class property. For example, the command generator **502** may detect a match of an event with the same room property as the device to which the command was sent. The command generator **502** may detect a match when a volume property is above a certain threshold.

If no match is detected, at step **1103** the command generator **502** may determine that the instructed device **306** is off, set to the wrong input, muted, or has its volume turned down. Accordingly, the command generator **502** may send one or more commands to turn the device on, switch the input, turn up the volume, and/or turn off the mute before looping back to the start of the process. The command generator **502** may randomly choose one of the options on each successive iteration of the loop, or the command generator **502** may cycle through each option on successive iterations.

When a match is detected, the command generator **502** may proceed to decision **1104** to ensure the volume is in an appropriate range. When the volume property of the detected audio is above or below certain thresholds, the command generator **502** may send one or more volume down or up commands respectively in step **1105**. In step **1106**, the command generator **502** may then cause the audio analyzer **503** to analyze an audio signal received after the volume commands were transmitted before looping back to decision **1104**. When the audio signal is within the volume thresholds, the process successfully completes.

The monitoring and security system **300** may monitor video events detected from video signals received by video cameras **310**. The video analyzer **505** may detect video events based on changes in video signals, such as brightness, color, etc., in similar ways as described for decision **1005**. The video camera **310** and/or the video analyzer **505** may detect video events when a brightness, color, or other aspect of a video signal received by a video camera **310** changes by a threshold amount. In this way, for example, a video camera **310** pointed at a television or display device may recognize when the television or display device is turned on (e.g., based on the change in brightness, color, etc. associated with a displayed image). The video analyzer **505** may use classification, image analysis, and/or machine vision techniques to categorize such events based on the video signals, and store the analyzed events in the log **601** according to the process shown at FIG. **10**. For example, the video analyzer **505** may detect that lights were turned on or off based on changes in brightness of a video signal recorded by video camera **310**. The video analyzer **505** may detect that a television is tuned to a particular network and/or channel based on receiving video signals containing a network logo or other image element associated with a network and/or channel. The video analyzer **505** may detect that a particular program is being displayed based on recognizing images associated with the program (e.g., using a fingerprinting process to compare an image fingerprint to a database of image fingerprints associated with particular programs). Indications of corresponding events (e.g., “light on” or “light off” events, events indicating a particular network, channel, and/or program is being displayed, etc.) may be stored in the logs **601**, and the profile generator **504** may further detect groups **602** of events and habits **603** therefrom. The profile generator **504** may map such habits to

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commands for turning lights on and off, turning a television on and off, tuning to a particular channel, network, or program, and the like according to a command schedule 901 or dynamic events 902. Other monitoring devices that monitor a premises 302 may also be used.

The premises controller 301 may detect an event and/or determine information about an event by requesting information from a server associated with a user entertainment device 306. For example, a server at the local office 103 (e.g., app server 107) may offer an application programming interface (API) for accessing information about set top box 113. Accordingly, the premises controller 301 may request information about an event associated with the set top box 113 and/or a television 307 connected to the set top box 113 by sending an API request to the app server 107. The app server 107 may respond with an indication of a channel, network, and/or program currently tuned by the set top box 113, an on/off status of the set top box 113, and the like. Accordingly, the premises controller 301 may detect events and/or obtain additional information about detected and/or analyzed events by sending a request to a device such as an API server. The information obtained from the device may be stored in the logs 601, and the profile generator 504 may further detect groups 602 of events and habits 603 therefrom. The profile generator 504 may map such habits 603 to commands such as turning a television on and off, tuning to a particular channel, network, or program, and the like, according to a command schedule 901 or dynamic events 902.

The monitoring and security system 300 also monitors home automation events, which may be detected by the automation controller 507. For example, the automation controller 507 may communicate with one or more smart devices such as lights 308 and other smart devices such as smart doorbells, smart locks, smart appliances, etc. The automation controller 507, using home automation protocols or other network connectivity, may regularly poll connected smart devices to detect whether their state has changed, and detect events therefrom. For example, the automation controller 507 may detect an event whenever an appliance is turned on or off and/or otherwise changes state. Properties of an event detected by the automation controller 507 may include an on/off state of the device, a room of the device, an intensity of a light 308, and the like. The smart devices may send notifications of their current state and/or changed state to the automation controller 507.

Furthermore, the command generator 502 and/or automation controller 507 may be configured to send commands to smart devices including lights 308. The command schedule 901 or dynamic events 902 may contain commands based on detected events received from the automation controller 507. When a command is directed to a smart device, the command may be sent by the command generator 502, the automation controller 507, and/or some other component of the premises controller 301 or network controller 303.

The premises controller 301 may implement a “welcome” mode after a mode is changed from “away” to “stay” or the monitoring and security system 300 otherwise detects that a user has returned to a premises. According to a “welcome” mode, the premises controller 301 may be configured to continue simulating a user’s presence (e.g., using the command schedule 901 and/or dynamic events 902) for a limited time after the monitoring and security system 300 is switched out of an “away” mode. For example, for a five-minute period after switching out of an “away” mode, the premises controller 301 may continue to transmit commands according to a command schedule 901 and/or

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dynamic events 902, thus activating user entertainment devices 306 and/or other devices for a user who has just returned to the premises. After the expiration of the limited time, the premises controller 301 may continue operating according to the procedures associated with the “stay” mode as described herein.

One or more aspects of the disclosure may be embodied in computer-usable data and/or computer-executable instructions, such as in one or more program modules, executed by one or more computers or other devices. Generally, program modules include routines, programs, objects, components, data structures, etc. that perform particular tasks or implement particular abstract data types when executed by a processor in a computer or other data processing device. The computer executable instructions may be stored on one or more computer readable media such as a hard disk, optical disk, removable storage media, solid state memory, RAM, etc. The functionality of the program modules may be combined or distributed as desired in various embodiments. In addition, the functionality may be embodied in whole or in part in firmware or hardware equivalents such as integrated circuits, field programmable gate arrays (FPGA), application-specific integrated circuits (ASIC), and the like. Particular data structures may be used to more effectively implement one or more aspects of the invention, and such data structures are contemplated within the scope of computer executable instructions and computer-usable data described herein.

Aspects of the disclosure have been described in terms of examples. While exemplary systems and methods as described herein embodying various aspects of the present disclosure are shown, it will be understood by those skilled in the art, that the disclosure is not limited to these embodiments. Modifications may be made by those skilled in the art, particularly in light of the foregoing teachings. For example, each of the features of the aforementioned examples may be utilized alone or in combination or sub-combination with elements of the other examples. For example, any of the above described systems and methods or parts thereof may be combined with the other methods and systems or parts thereof described above. For example, the steps shown in the figures may be performed in other than the recited order, and one or more steps shown may be optional in accordance with aspects of the disclosure. It will also be appreciated and understood that modifications may be made without departing from the true spirit and scope of the present disclosure. The description is thus to be regarded as illustrative instead of restrictive on the present disclosure.

The invention claimed is:

1. A method comprising:

detecting, by a computing device and while one or more users are present at a premises, an event pair pattern in which a second event occurs following occurrence of a first event, wherein the second event comprises light or sound;  
detecting the first event while the one or more users are not present at the premises; and  
sending, based on the detection of the first event and the second event not occurring after the detection, one or more commands to cause control of one or more light or sound sources to simulate the second event.

2. The method of claim 1, wherein the detecting the event pair pattern comprises:

determining the first event as a causal event; and  
determining the second event as a reaction event that occurs based on the causal event.



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3. The method of claim 1, wherein:  
the first event comprises a doorbell ringing;  
the second event comprises a dog barking; and  
the sending the one or more commands causes output of  
sound of the dog barking based on the doorbell ringing. 5
4. The method of claim 1, further comprising:  
detecting, after the sending the one or more commands,  
light or sound emitted from the one or more sources;  
and  
determining, based on determining that the light or sound 10  
emitted from the one or more sources matches the  
second event, that the one or more commands were  
successfully executed.
5. The method of claim 1, further comprising:  
detecting, after the sending the one or more commands, 15  
light or sound emitted from the one or more light or  
sound sources; and  
sending, based on determining that the light or sound  
emitted from the one or more light or sound sources  
does not match the second event, one or more second 20  
commands.
6. The method of claim 1, wherein the sending the one or  
more commands comprises instructing an audio output  
device to output an audio signal recording associated with  
the second event. 25
7. A computing device comprising:  
one or more processors;  
memory storing instructions that, when executed by the  
one or more processors, configure the computing  
device to:  
detect, while one or more users are present at a prem-  
ises, an event pair pattern in which a second event  
occurs following occurrence of a first event, wherein  
the second event comprises light or sound;  
detecting the first event while the one or more users are 35  
not present at the premises; and  
sending, based on the detection of the first event and the  
second event not occurring after the detection, one or  
more commands to cause control of one or more  
light or sound sources to simulate the second event. 40
8. The computing device of claim 7, wherein the instruc-  
tions, when executed by the one or more processors, con-  
figure the computing device to detect the event pair pattern  
by:  
determining the first event as a causal event; and  
determining the second event as a reaction event that  
occurs based on the causal event.
9. The computing device of claim 7, wherein:  
the first event comprises a doorbell ringing;  
the second event comprises a dog barking; and  
the instructions, when executed by the one or more  
processors, configure the computing device to send the  
one or more commands causes output of sound of the  
dog barking based on the doorbell ringing.
10. The computing device of claim 7, wherein the instruc- 55  
tions, when executed by the one or more processors, further  
configure the computing device to:  
detecting, after the sending the one or more commands,  
light or sound emitted from the one or more sources;  
and  
determining, based on determining that the light or sound  
emitted from the one or more sources matches the  
second event, that the one or more commands were  
successfully executed. 60

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11. The computing device of claim 7, wherein the instruc-  
tions, when executed by the one or more processors, further  
configure the computing device to:  
detecting, after the sending the one or more commands,  
light or sound emitted from the one or more light or  
sound sources; and  
sending, based on determining that the light or sound  
emitted from the one or more light or sound sources  
does not match the second event, one or more second  
commands.
12. The computing device of claim 7, wherein the instruc-  
tions, when executed by the one or more processors, con-  
figure the computing device to send the one or more com-  
mands comprises instructing an audio output device to  
output an audio signal recording associated with the second  
event.
13. At least one non-transitory computer readable storage  
medium storing computer readable instructions which, when  
executed, cause performance of actions comprising:  
detecting, while one or more users are present at a  
premises, an event pair pattern in which a second event  
occurs following occurrence of a first event, wherein  
the second event comprises light or sound;  
detecting the first event while the one or more users are  
not present at the premises; and  
sending, based on the detection of the first event and the  
second event not occurring after the detection, one or  
more commands to cause control of one or more light  
or sound sources to simulate the second event.
14. The at least one non-transitory computer readable  
storage medium of claim 13, wherein the instructions, when  
executed, cause detecting the event pair pattern by cause  
performance of actions comprising:  
determining the first event as a causal event; and  
determining the second event as a reaction event that  
occurs based on the causal event.
15. The at least one non-transitory computer readable  
storage medium of claim 13, wherein:  
the first event comprises a doorbell ringing;  
the second event comprises a dog barking; and  
the instructions, when executed, cause sending the one or  
more commands causes output of sound of the dog  
barking based on the doorbell ringing.
16. The at least one non-transitory computer readable  
storage medium of claim 13, wherein the instructions, when  
executed, further cause performance of actions comprising:  
detecting, after the sending the one or more commands,  
light or sound emitted from the one or more sources;  
and  
determining, based on determining that the light or sound  
emitted from the one or more sources matches the  
second event, that the one or more commands were  
successfully executed.
17. The at least one non-transitory computer readable  
storage medium of claim 13, wherein the instructions, when  
executed, further cause performance of actions comprising:  
detecting, after the sending the one or more commands,  
light or sound emitted from the one or more light or  
sound sources; and  
sending, based on determining that the light or sound  
emitted from the one or more light or sound sources  
does not match the second event, one or more second  
commands.